DataBase Lab 07 (In-Lab Task)

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Task 1

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Worksheet | Query Builder

```
SELECT e.first_name, e.job_id, d.department_name, e.salary
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
ORDER BY d.department_name ASC;
```

Query Result

SQL | Fetched 50 rows in 0.004 seconds

	FIRST_NAME	JOB_ID	DEPARTMENT_NAME	SALARY
1	Shelley	AC_MGR	Accounting	12008
2	William	AC_ACCOUNT	Accounting	8300
3	Jennifer	AD_ASST	Administration	4400
4	Lex	AD_VP	Executive	17000
5	Steven	AD_PRES	Executive	24300
6	Neena	AD_VP	Executive	17000
7	John	FI_ACCOUNT	Finance	8200
8	Daniel	FI_ACCOUNT	Finance	9000
9	Jose Manuel	FI_ACCOUNT	Finance	7800
10	Ismael	FI_ACCOUNT	Finance	7700
11	Luis	FI_ACCOUNT	Finance	6900
12	Nancy	FI_MGR	Finance	12008
13	Susan	HR_REP	Human Resources	6500
14	Diana	IT_PROG	IT	4200
15	Valli	IT_PROG	IT	4800
16	Bruce	IT_PROG	IT	6000
17	Mujtaba	IT_PROG	IT	60000
18	Alexander	IT_PROG	IT	9000
19	David	IT_PROG	IT	4800
20	Pat	MK_REP	Marketing	6000
21	Michael	MK_MAN	Marketing	13000
22	Hermann	PR_REP	Public Relations	10000
23	Guy	PU_CLERK	Purchasing	2600
24	Sigal	PU_CLERK	Purchasing	2800
25	Alexander	PU_CLERK	Purchasing	3100
26	Den	PU_MAN	Purchasing	11000
27	Karen	PU_CLERK	Purchasing	2500
28	Shelli	PU_CLERK	Purchasing	2900
29	Mattea	SA_REP	Sales	7200
30	David	SA_REP	Sales	6800
31	Sundar	SA_REP	Sales	6400
32	Amit	SA_REP	Sales	6200

Task 2

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Worksheet | Query Builder

```
SELECT department_id, COUNT(*) AS employee_count
FROM employees
GROUP BY department_id
HAVING COUNT(*) >= 2;
```

Query Result

SQL | All Rows Fetched: 9 in 0.002 seconds

	DEPARTMENT_ID	EMPLOYEE_COUNT
1	100	6
2	30	6
3	(null)	2
4	90	3
5	20	2
6	110	2
7	50	45
8	80	34
9	60	6

Task 3

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Worksheet | Query Builder

```
SELECT e.first_name, e.job_id,
       e.salary*12 AS annual_salary,
       d.department_name, l.city
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
JOIN locations l
ON d.location_id = l.location_id
WHERE e.salary*12 >= 60000
AND e.job_id <> 'ANALYST';
```

Query Result

SQL | Fetched 50 rows in 0.006 seconds

	FIRST_NAME	JOB_ID	ANNUAL_SALARY	DEPARTMENT_NAME	CITY
1	Bruce	IT_PROG	72000	IT	Southlake
2	Mujtaba	IT_PROG	720000	IT	Southlake
3	Alexander	IT_PROG	108000	IT	Southlake
4	Adam	ST_MAN	98400	Shipping	South San Francisco
5	Kevin	ST_MAN	69600	Shipping	South San Francisco
6	Shanta	ST_MAN	78000	Shipping	South San Francisco
7	Payam	ST_MAN	94800	Shipping	South San Francisco
8	Matthew	ST_MAN	96000	Shipping	South San Francisco
9	William	AC_ACCOUNT	99600	Accounting	Seattle
10	Shelley	AC_MGR	144096	Accounting	Seattle
11	John	FI_ACCOUNT	98400	Finance	Seattle
12	Ismael	FI_ACCOUNT	92400	Finance	Seattle
13	Jose Manuel	FI_ACCOUNT	93600	Finance	Seattle
14	Luis	FI_ACCOUNT	82800	Finance	Seattle
15	Nancy	FI_MGR	144096	Finance	Seattle
16	Daniel	FI_ACCOUNT	108000	Finance	Seattle
17	Neena	AD_VP	204000	Executive	Seattle
18	Steven	AD_PRES	291600	Executive	Seattle
19	Lex	AD_VP	204000	Executive	Seattle
20	Den	PU_MAN	132000	Purchasing	Seattle
21	Michael	MK_MAN	156000	Marketing	Toronto
22	Pat	MK_REP	72000	Marketing	Toronto
23	Susan	HR_REP	78000	Human Resources	London
24	Alyssa	SA_REP	105600	Sales	Oxford
25	Jonathon	SA_REP	103200	Sales	Oxford
26	Ellen	SA_REP	132000	Sales	Oxford
27	Charles	SA_REP	74400	Sales	Oxford
28	Jack	SA_REP	100800	Sales	Oxford
29	John	SA_MAN	168000	Sales	Oxford

Task 4

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Worksheet | Query Builder

```
SELECT DISTINCT
       M.EMPLOYEE_ID,
       M.FIRST_NAME,
       M.LAST_NAME
FROM EMPLOYEES E
JOIN EMPLOYEES M
ON E.MANAGER_ID = M.EMPLOYEE_ID;
```

Query Result

SQL | All Rows Fetched: 18 in 0.006 seconds

	EMPLOYEE_ID	FIRST_NAME	LAST_NAME
1	101	Neena	Kochhar
2	108	Nancy	Greenberg
3	147	Alberto	Errazuriz
4	205	Shelley	Higgins
5	102	Lex	De Haan
6	120	Matthew	Weiss
7	124	Kevin	Mourgos
8	148	Gerald	Cambrault
9	201	Michael	Hartstein
10	100	Steven	King
11	114	Den	Raphaely
12	121	Adam	Fripp
13	123	Shanta	Vollman
14	145	John	Russell
15	122	Payam	Kaufling
16	146	Karen	Partners
17	149	Eleni	Zlotkey
18	103	Alexander	Hunold

Task 5

Worksheet | Query Builder

```
SELECT d.department_id, d.department_name
FROM departments d
LEFT JOIN employees e
ON d.department_id = e.department_id
WHERE e.employee_id IS NULL;
```

Query Result

SQL | All Rows Fetched: 16 in 0.002 seconds

	DEPARTMENT_ID	DEPARTMENT_NAME
1	120	Treasury
2	130	Corporate Tax
3	140	Control And Credit
4	150	Shareholder Services
5	160	Benefits
6	170	Manufacturing
7	180	Construction
8	190	Contracting
9	200	Operations
10	210	IT Support
11	220	NOC
12	230	IT Helpdesk
13	240	Government Sales
14	250	Retail Sales
15	260	Recruiting
16	270	Payroll

Task 6

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Worksheet | Query Builder

```
SELECT e.first_name, e.salary, d.department_name
FROM employees e
LEFT JOIN departments d
ON e.department_id = d.department_id;
```

Query Result

SQL | Fetched 50 rows in 0.003 seconds

	FIRST_NAME	SALARY	DEPARTMENT_NAME
1	Jennifer	4400	Administration
2	Pat	6000	Marketing
3	Michael	13000	Marketing
4	Karen	2500	Purchasing
5	Guy	2600	Purchasing
6	Sigal	2800	Purchasing
7	Shelli	2900	Purchasing
8	Alexander	3100	Purchasing
9	Den	11000	Purchasing
10	Susan	6500	Human Resources
11	Douglas	2600	Shipping
12	Donald	2600	Shipping
13	Kevin	3000	Shipping
14	Alana	3100	Shipping
15	Vance	2800	Shipping
16	Samuel	3200	Shipping
17	Britney	3900	Shipping
18	Sarah	4000	Shipping
19	Randall	2500	Shipping
20	Timothy	2900	Shipping
21	Jennifer	3600	Shipping
22	Kelly	3800	Shipping
23	Anthony	3000	Shipping
24	Julia	3400	Shipping
25	Alexis	4100	Shipping
26	Nandita	4200	Shipping
27	Girard	2800	Shipping
28	Martha	2500	Shipping
29	Jean	3100	Shipping
30	Winston	3200	Shipping
31	Peter	2500	Shipping

Task 7

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Worksheet | Query Builder

```
SELECT e.first_name, e.job_id, d.department_name, l.city
FROM employees e
JOIN departments d
ON e.department_id = d.department_id
JOIN locations l
ON d.location_id = l.location_id
WHERE l.state_province IS NULL;
```

Query Result

SQL | All Rows Fetched: 1 in 0.005 seconds

	FIRST_NAME	JOB_ID	DEPARTMENT_NAME	CITY
1	Susan	HR_REP	Human Resources	London

Task 8

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Worksheet | Query Builder

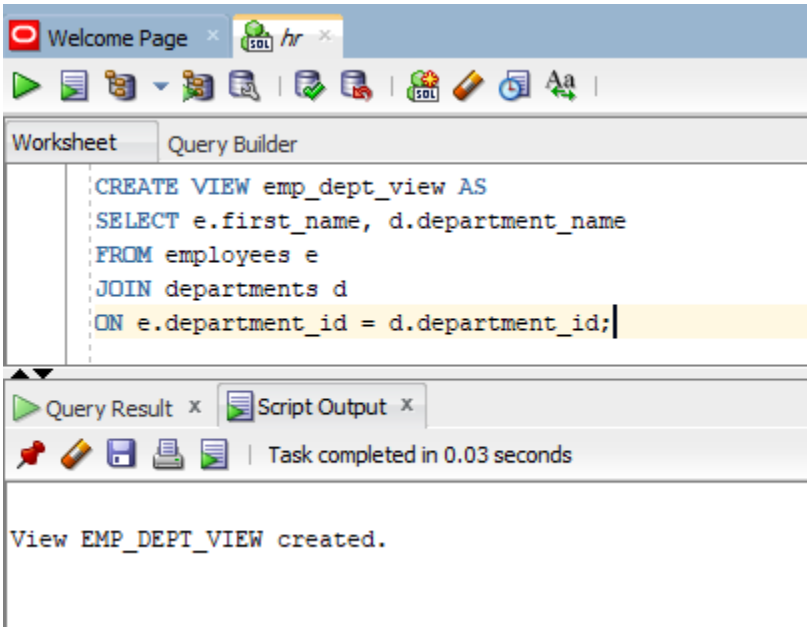
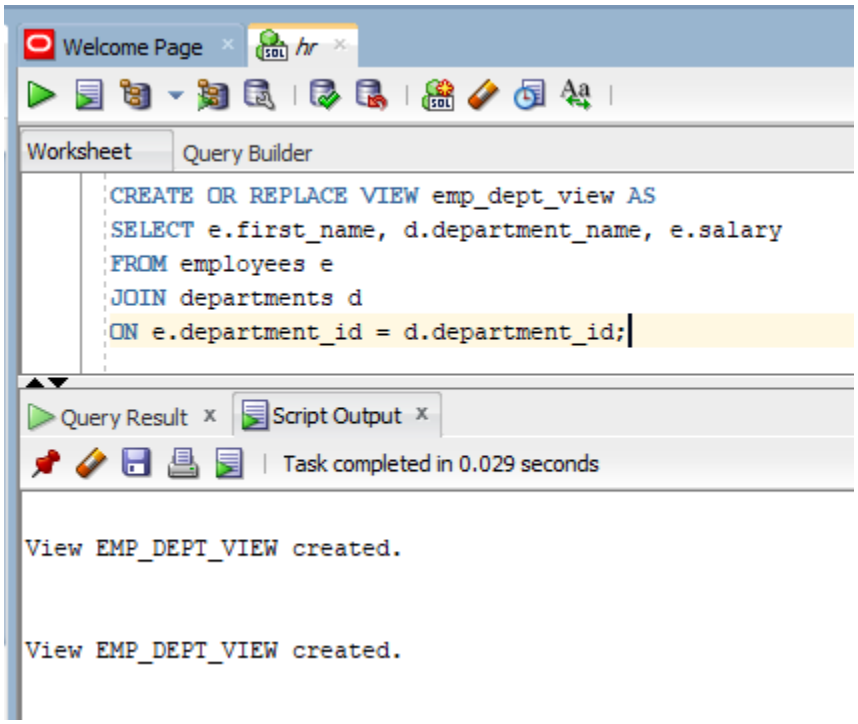
```
SELECT employee_id
FROM employees
MINUS
SELECT employee_id
FROM job_history;
```

Query Result

SQL | Fetched 50 rows in 0.005 seconds

	EMPLOYEE_ID
1	55
2	100
3	106
4	107
5	108
6	109
7	110
8	111
9	112
10	113
11	115
12	116
13	117
14	118
15	119
16	120
17	121
18	123
19	124
20	125
21	126
22	127
23	128
24	129
25	130
26	131
27	132
28	133
29	134
30	135

Task 9



Task 10